Analysis of Retinal Representations in Vision Transformers

Stanford CS375N Final Project

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Abstract

The primate visual system processes information hierarchically, beginning at the retina and passing through subcortical structures before reaching the primary visual cortex (V1). While deep learning models have shown remarkable success in matching cortical representations, their alignment with early, pre-cortical visual processing stages remains less understood. This study investigates the representational alignment between a biologically-constrained Primate Retina Convolutional Neural Network (prCNN) and the self-supervised Vision Transformer, DINOv2. Two central research questions are explored: 1) How much subcortical processing occurs between the retina and V1? and 2) Where does DINOv2 encode retinal-like features? By evaluating these models against `TVSDV1` fMRI recordings using cross-validated ridge regression and variance partitioning, this study finds that DINOv2 encoder 2 achieves the best V1 match (ceiled $R^2 \approx 0.72$) and vastly outperforms the prCNN (`conv1` ceiled $R^2 \approx 0.24$), indicating a substantial computational role for subcortical processing; variance partitioning confirms that DINOv2 accounts for unique V1 variance in over 91% of voxels. Using Linear Centered Kernel Alignment (CKA), cross-validated ridge regression, and Representational Similarity Analysis (RSA, Spearman r) for direct cross-model comparison on `TVSDStimulusTrainSet`, this study reveals that DINOv2 encodes retinal-like features most strongly in its initial embeddings layer and the first transformer block (encoder 0), with similarity dropping sharply in deeper blocks. These findings highlight the emergence of biologically plausible early visual representations in self-supervised transformers, while also underscoring the substantial subcortical transformation that separates retinal from cortical processing.

1 Introduction

A central goal in computational neuroscience is to build models that not only functionally parallel biological neural structures but also accurately predict biological neural responses. The biological visual pathway begins at the retina, where photoreceptors capture light, bipolar cells process it, and retinal ganglion cells (RGCs) transmit the signals via the optic nerve to the lateral geniculate nucleus (LGN) and subsequently to the primary visual cortex (V1).

Historically, Convolutional Neural Networks (CNNs) have been the standard for modeling the visual cortex, with early layers mapping to V1 and deeper layers mapping to inferior temporal (IT) cortex. Recently, Vision Transformers (ViTs) trained with self-supervised learning, such as DINOv2, have demonstrated highly robust and biologically plausible representations [3]. However, fewer studies have explicitly analysed how these large-scale foundation models encode subcortical neural structures like the retina.

This study investigates the representational similarities between DINOv2 and a biologically-inspired primate retina CNN (prCNN) that is explicitly trained to predict the firing rates of macaque RGCs [2]. With BBScore [1], this study seeks to address two key questions through this comparison:

1. **How much subcortical processing is there between the retina and V1?** By comparing how well the prCNN and DINOv2 layers predict V1 fMRI activity, this study can infer the extent of visual information processing that occurs after the retina and before V1.
2. **Where does DINOv2 encode for the retina?** By directly comparing the internal representations of the prCNN and DINOv2, this study aims to locate where retinal-like features emerge within the transformer architecture.

2 Related Works

Prior research has extensively compared deep neural networks to the primate cortical visual system. Self-supervised Vision Transformers like DINOv2 [3] have been shown to learn powerful visual features that achieve state-of-the-art alignment with V1 and higher visual areas. For the retina, CNN models have recently superseded traditional Linear-Nonlinear (LN) cascade models for accurately capturing the firing rates of numerically dominant RGC types in the macaque and human retina given visual stimuli [2].

This study contributes to the BBScore framework [1] in the following ways [1]:

- **Model Integration**: This study introduced the biologically-constrained prCNN into the BBScore registry, enabling standardized feature extraction and evaluation.
- **Benchmark Extension**: This study created modular scripts for variance partitioning and cross-model alignment (CKA, RSA, and ridge regression) that extend the core capabilities of BBScore for model-to-model investigations.
- **Pipeline Optimization**: This study optimized the analysis pipeline to slice input data based on available system RAM and VRAM before loading, avoiding out-of-memory errors.

3 Approach

The core hypothesis is that DINOv2 naturally learns early visual representations that parallel the primate retina in its initial layers, and that the pathway from retina to V1 involves significant processing that DINOv2 captures better than a retina-only model. This study expects that DINOv2 will outperform the prCNN in predicting V1 responses, indicating a substantial role for subcortical processing. Furthermore, this study expects that the earliest layers of DINOv2 will exhibit the highest representational similarity to the prCNN, reflecting the hierarchical nature of visual processing.

To test this, this study uses the `TVSDV1` dataset for V1 matching and the `TVSDStimulusTrainSet` for cross-model analysis. This study compares the prCNN (a biologically-constrained model) and DINOv2 (a self-supervised Vision Transformer), alongside baseline raw pixels as the baseline. The comparison methods include cross-validated ridge regression, variance partitioning, Linear Centered Kernel Alignment (CKA), and Representational Similarity Analysis (RSA). This experimental design allows this study to both quantify the alignment of each model with biological V1 and directly map the internal representations of the neural network models to each other.

4 Experiments

4.1 Data

For the model to V1 match analysis, this study uses the `TVSDV1` dataset: fMRI recordings from the primary visual cortex of primates viewing natural images, which this study treats as biological ground truth for early cortical visual processing. The stimuli with which the fMRI recordings were obtained, `TVSDStimulusTrainSet`, are provided to each model as input. Stimulus preprocessing follows each model's training regime: The prCNN was trained on grayscale images each jittered across 50 frames to approximate saccades; this study therefore converts each input image to grayscale and repeats it across 50 frames. DINOv2 was trained on full-color `TVSD` images; thus this study evaluates it on both grayscale and full-color stimuli—the former controls for chromatic information in the input, and the latter matches the model's native, color-trained operating regime.

[1] The codebase is available at `https://github.com/briansebzhou/bbscore_public`.

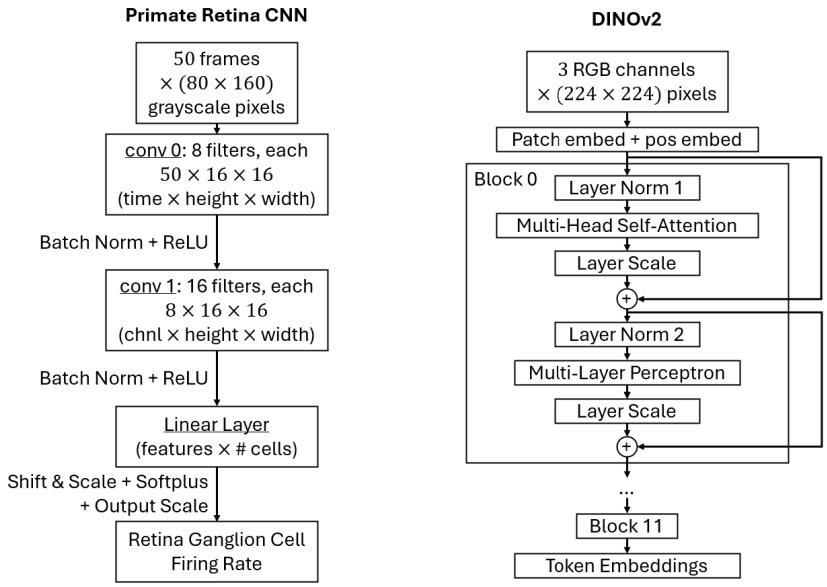

Figure 1: Schematics of the prCNN (left) and DINOv2 ViT-Base (right): inputs, main layers, and outputs for each model used for analysis in BBScore.

For the cross-model analysis, the features of each model layer produced in the model to V1 analysis are reused and compared against each other.

4.2 Models

prCNN: The prCNN model is trained on RGC responses to natural images [2]. The model takes as input a stimulus movie consisting of 50 frames of a grayscale image jittered over time to approximate saccades. The architecture consists of:

- **conv0**: The first layer applies eight convolution filters over the 50 temporal frames (treated as input channels) followed by Batch Normalization and a ReLU activation.
- **conv1**: The second layer applies sixteen convolution filters followed by Batch Normalization and ReLU.
- **linear**: A fully-connected linear layer followed by cell-specific shift/scale parameters and a softplus nonlinearity to output the firing rates of specific RGC types.

DINOv2: The ViT-Base variant of DINOv2 [3] is used, which processes static RGB images divided into patches. The architecture consists of a patch embedding layer followed by 12 transformer blocks. Each block contains four main sublayers: Layer Norm 1 (`norm1`), Multi-Head Self-Attention (`attention`), Layer Norm 2 (`norm2`), and a Multi-Layer Perceptron (`mlp`). We also evaluate a grayscale variant of DINOv2 to control for color as a confounding variable.

Figure 1 summarizes the two model architectures alongside the layer naming used in this study.

Baselines: This study includes two raw pixel baselines to establish lower bounds for V1 prediction: a grayscale luminance baseline (preprocessed to match retina model inputs) and a high-resolution RGB baseline (raw images matching DINOv2 inputs) to isolate the contribution of learned features in each model.

4.3 Evaluation Method

This study's approach consists of two main analyses:

1. V1 Match Analysis: The representations of the layers of each model are mapped to V1 neural recordings to determine which model better captures V1 variance. Cross-validated ridge regression is used to predict voxel responses from model features. To disentangle the contributions of each

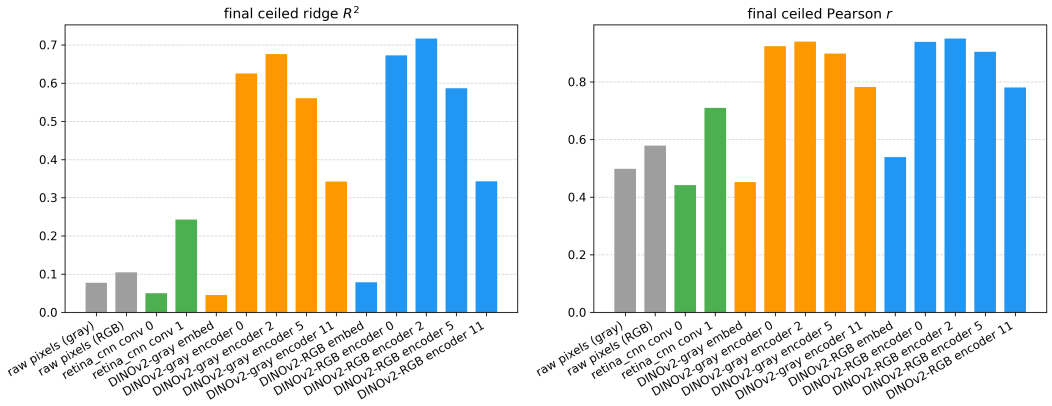

Figure 2: Ceiled ridge R^2 (left) and ceiled Pearson r (right) for V1 fMRI prediction, aggregated across voxels, for raw pixels, prCNN layers, and DINOv2-gray and DINOv2-RGB embeddings and encoder layers.

model, variance partitioning is performed: the best layers of each model are combined into a joint layer, for which the joint R^2 is calculated, and the unique variance of each model layer is obtained by subtracting the other model's R^2 from the joint R^2 (e.g. $\text{Unique}_{\text{DINOv2}} = R^2_{\text{Joint}} - R^2_{\text{Retina}}$).

2. Cross-Model Analysis: This study directly compares the layer-wise activations of the prCNN (`conv0` and `conv1`) with the encoder blocks and encoder sublayers of DINOv2 using three complementary approaches: cross-validated ridge regression, which quantifies how well activations in one model linearly predict those in the other; Linear Centered Kernel Alignment (CKA), which measures representational similarity while remaining invariant to orthogonal transformations; and Representational Similarity Analysis (RSA) via Spearman correlation between the prCNN and DINOv2 Representational Dissimilarity Matrices (RDMs). Prior to computing CKA, RSA, and ridge regression for cross-model analysis, representations are reduced to 1000 components using Principal Component Analysis (PCA).

5 Results

5.1 Step 1: Match to V1 (Subcortical Processing)

This study's analysis reveals that DINOv2 matches V1 responses significantly better than the prCNN. As shown in Figure 2, the overall metrics show that prCNN `conv0` ($R^2 = 0.050$, Pearson $r = 0.44$) and the initial embeddings layer of DINOv2 ($R^2 = 0.079$, $r = 0.54$ for RGB; $R^2 = 0.045$, $r = 0.45$ for grayscale) perform similarly to raw pixels in V1 alignment (grayscale pixel: $R^2 = 0.078$, $r = 0.50$; RGB pixel: $R^2 = 0.105$, $r = 0.58$). However, prCNN `conv1` shows much better V1 alignment than `conv0` ($R^2 = 0.243$, $r = 0.71$), and all four encoder layers of DINOv2 investigated here show significantly better V1 alignment than prCNN `conv1`. DINOv2-RGB encoder 2 achieves the best overall V1 match ($R^2 = 0.717$, $r = 0.950$), representing a nearly $3\times$ improvement in R^2 over the best prCNN layer. DINOv2-gray encoder 2 is the best grayscale layer ($R^2 = 0.676$, $r = 0.940$). Both encoder 5 and encoder 11 show declining but still substantially higher V1 alignment than the prCNN (encoder 5: $R^2 \approx 0.57$–0.59; encoder 11: $R^2 \approx 0.34$). Furthermore, looking at the two raw pixel baselines, color information contributes a nontrivial level of alignment to V1 (a $\Delta R^2 \approx 0.03$ difference between grayscale and RGB pixels). When comparing the layers of DINOv2 given grayscale stimuli versus RGB stimuli, DINOv2 given RGB stimuli performs marginally better ($\Delta R^2 \approx 0.04$ at encoder 2), though both vastly outperform the prCNN. Complementary RSA analysis confirms that these improvements are not solely due to improved regression fit: DINOv2 encoder 2 also achieves higher V1 RSA Spearman r ($r \approx 0.13$–0.14) than both raw pixels and prCNN `conv1` ($r \approx 0.06$), though all RSA values in this V1 setting remain modest in absolute terms. Note that for these overall metrics, the noise ceiling is incorporated to account for the inherent noise in neural recordings, isolating how well the model layers can align with V1.

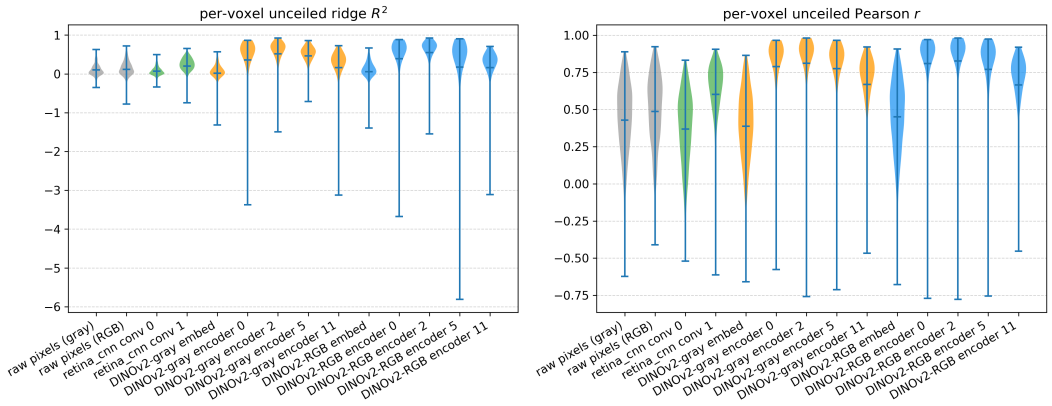

Figure 3: Per-voxel distributions of unceiled ridge R^2 (left) and unceiled Pearson r (right) for V1 fMRI prediction, across raw pixels, prCNN (`conv0`, `conv1`), and DINOv2-gray and DINOv2-RGB embeddings and encoder layers.

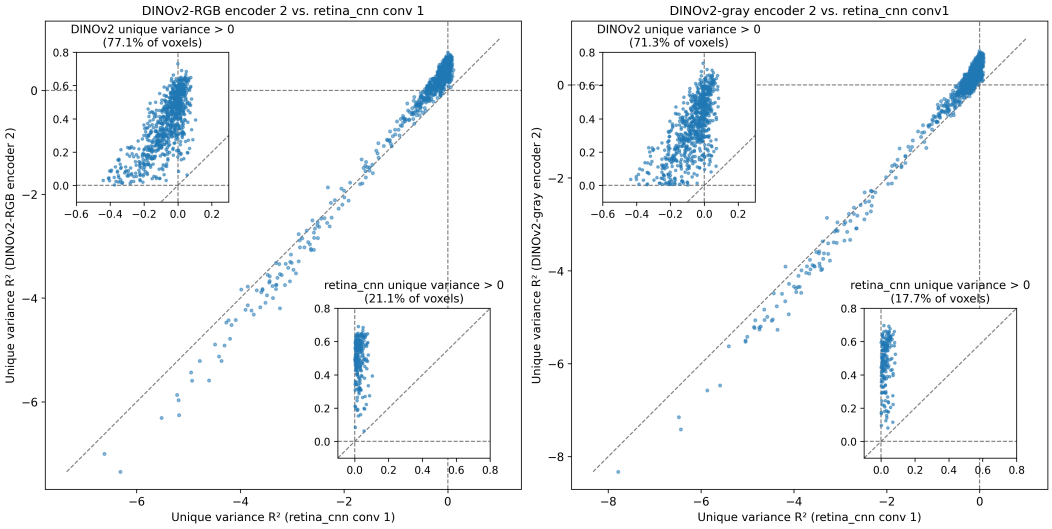

Figure 4: Per-voxel unique variance (R^2) from variance partitioning: prCNN (`conv1`) versus DINOv2-RGB encoder 2 (left) and DINOv2-gray encoder 2 (right), with identity and zero-reference lines and insets for voxels with positive unique variance on each axis.

Figure 3 illustrates this advantage at a more granular, per-voxel level. It can be observed that better-matching layers have less spread in their distributions than worse-matching layers. This tighter spread indicates that layers with better mean alignment for V1 match V1 more consistently across the entire recording region. Note that unceiled metrics are used for this per-voxel analysis because a few voxels have noise ceilings so low that ceiling correction causes numerical instability, which would artificially widen the spread of datapoints too much and obscure clarity for analysis.

The variance partitioning analysis (Figure 4) extends the per-voxel analysis to identify if there are certain voxels for which the prCNN predicts better than DINOv2 and vice versa. The results show that DINOv2 outperforms the prCNN across almost all voxels. DINOv2-RGB encoder 2 versus prCNN `conv1` yields 91.7% of voxels with higher unique variance for DINOv2; DINOv2-gray encoder 2 yields 91.1%. Focusing specifically on the voxels for which the unique variance of each model is positive, literally all of them have higher unique variance for DINOv2 than the prCNN. This further strengthens the argument that DINOv2 is better than the prCNN for aligning with V1 across the entire cortical region.

5.2 Step 2: Cross-Model Analysis (Retinal Encoding in DINOv2)

When directly comparing the models using Linear CKA, ridge regression, and RSA (Spearman r between cross-model RDMs), this study finds that the "retina" in DINOv2 corresponds to its very initial processing stages. Specifically, as shown in Figure 5, a broad coarse sweep across all of DINOv2 reveals that encoder 0 is almost always the best retina-matching layer, regardless of the metric analyzed (with the minor exception of `conv0` versus DINOv2-gray on RSA, where the preceding embeddings layer performs best with $r = 0.44$). For `conv0` versus DINOv2-RGB, encoder 0 attains CKA = 0.31, RSA r = 0.32, and ridge R^2 = 0.29–0.30 (retina→DINOv2 and DINOv2→retina respectively), compared to CKA $\leq$ 0.21 and RSA $r \leq$ 0.21 for encoders 2 and deeper. For the same prCNN layer, DINOv2 given grayscale stimuli shows a better match than DINOv2 given RGB stimuli (e.g. RSA r = 0.44 vs. 0.32 for `conv0` at the best encoder-0/embeddings layer), which is expected since the prCNN is trained exclusively on grayscale images. Furthermore, for the same DINOv2 layer, `conv0` matches better than `conv1` (e.g. CKA = 0.31 vs. 0.26 at encoder 0 for RGB). This is likely because `conv0` is closer to the raw input images that DINOv2 is more familiar with, whereas `conv1` is more specialized to retinal activity that DINOv2 was never exposed to. For both retinal layers, similarity drops off sharply in subsequent transformer blocks—by encoder 7, CKA falls to ≈ 0.09 and ridge R^2 drops to near zero, and by encoders 8–11 all metrics approach zero or become negative—indicating that retinal features are rapidly transformed into higher-level representations early in the network of DINOv2.

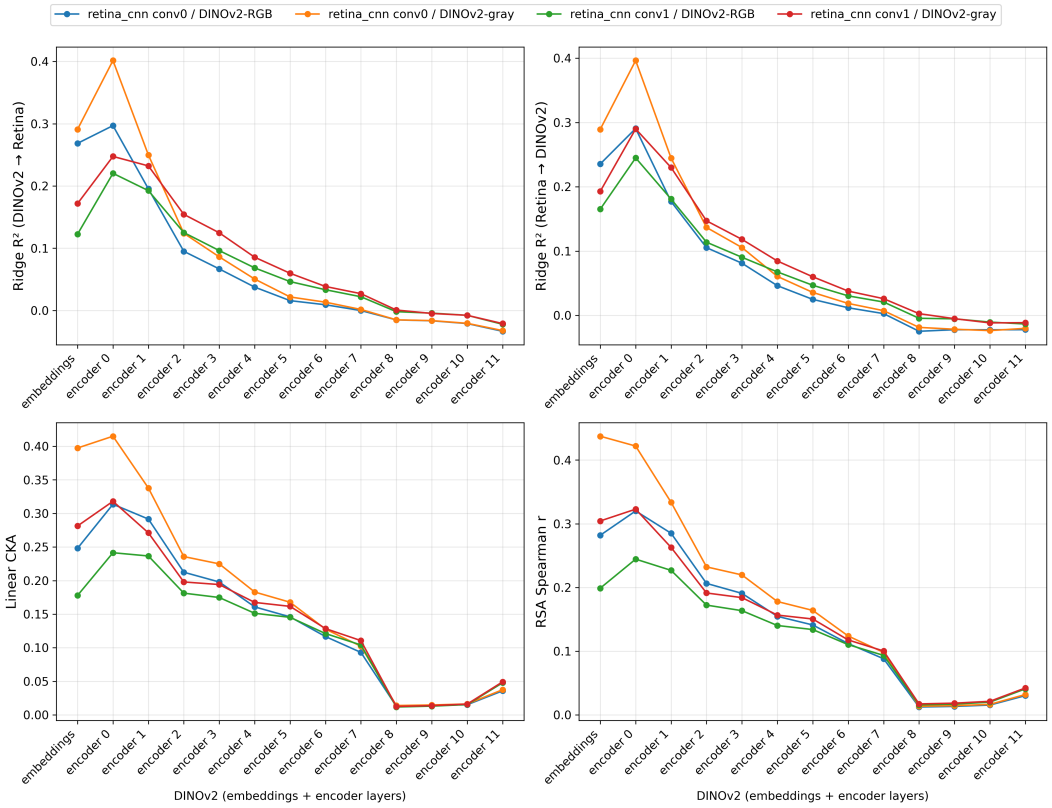

Figure 5: Cross-model metrics between prCNN `conv0`/`conv1` and DINOv2 (embeddings through encoder 11): ridge R^2 with DINOv2 predicting retina (top left) and retina predicting DINOv2 (top right), linear CKA (bottom left), and RSA Spearman r (bottom right). Curves compare RGB and grayscale DINOv2 for each retinal layer.

Having identified encoder 0 as the best match to the retina, this study takes a closer look at its sublayers (alongside the preceding embeddings layer) to find exactly where the retina is best represented. Figure 6 reports the corresponding metrics.

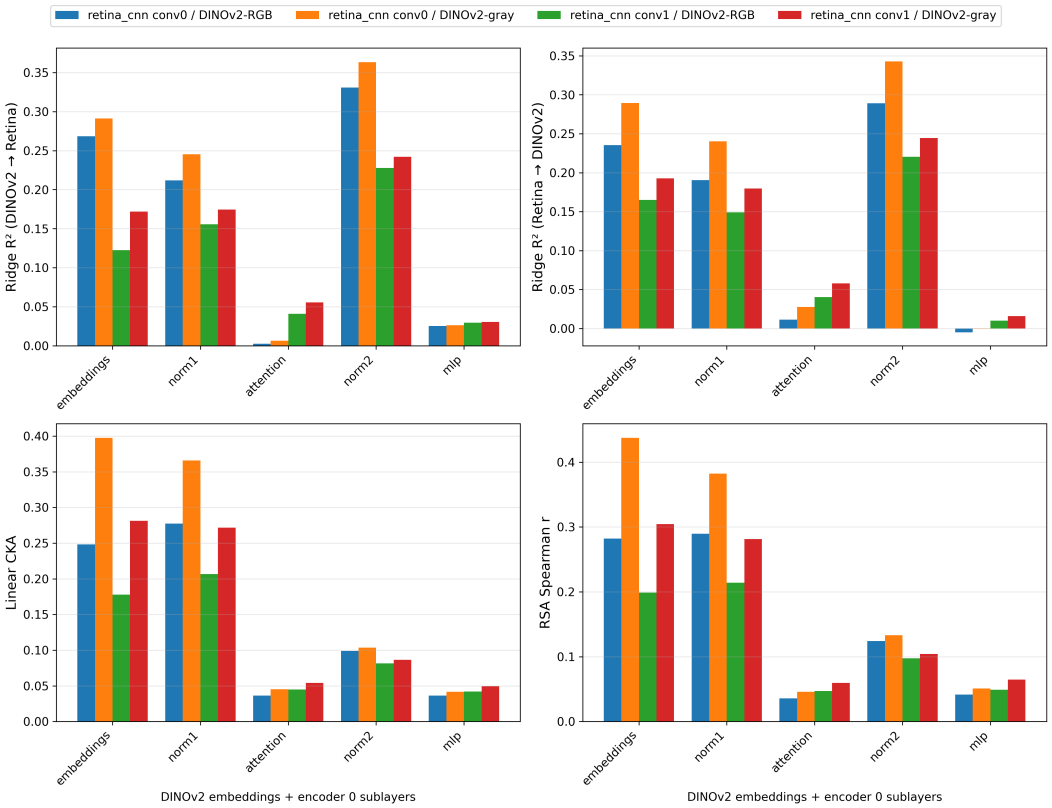

Figure 6: Bar charts of ridge R^2 (DINOv2 $\rightarrow$ Retina, and Retina $\rightarrow$ DINOv2), linear CKA, and RSA Spearman r between prCNN `conv0`/`conv1` and DINOv2 embeddings layer and encoder block 0 sublayers (`norm1`, `attention`, `norm2`, `mlp`), for RGB and grayscale DINOv2.

DINOv2-gray matches better across all sublayer conditions. For the retina layers, `conv0` matches better for embeddings and norm sublayers, while `conv1` matches relatively better for `attention` and `mlp` sublayers. Embeddings and `norm1` are nearly identical across all four metrics, proving that `norm1` is effectively a pass-through. `conv0` matches embeddings slightly better, while `conv1` matches `norm1` slightly better; however, the gap between embeddings and `norm1` is small compared to the drop-off to `attention` and `mlp` (and also `norm2` in the case of Linear CKA and RSA Spearman r). For example, for `conv0` vs. DINOv2-gray, the embeddings and `norm1` share nearly the same CKA (≈ 0.37 and RSA $r \approx 0.44$), while `attention` drops to CKA ≈ 0.26 and `mlp` to CKA ≈ 0.24. For Ridge R^2, `norm2` is the highest-matching sublayer overall; for Linear CKA and RSA Spearman r, however, `norm2` drops to near `attention` and `mlp` levels. This divergence occurs because `norm2` contains both residual (embedding) and attention information; Ridge R^2 benefits from the richer feature set, while CKA and RSA penalize the structural distortion introduced by attention.

6 Analysis

The results strongly support this study's hypotheses, revealing distinct insights into both the computational gap between the retina and V1, and the emergence of retinal-like representations in self-supervised vision transformers.

6.1 Step 1: V1 Match Alignment

The finding that DINOv2 vastly outperforms the prCNN in predicting V1 activity indicates that subcortical processing—from the retina, through the LGN, to V1—plays a highly significant role in transforming visual representations. The prCNN's `conv0` maps structurally to photoreceptors and bipolar cells, which perform minimal processing of raw pixels. Meanwhile, `conv1` maps to RGCs, which possess receptive fields and perform preliminary local visual processing. However, due to

the simplicity of the biological retina (and the corresponding prCNN), visual information is only processed very locally, falling far short of the sophistication of V1.

In contrast, DINOv2 encoder 2 best represents V1, and all four investigated DINOv2 encoder layers outperform all prCNN layers. This is likely because every transformer encoder layer contains attention mechanisms that integrate information globally across the entire visual field. This global integration is a much better match for the higher-level cortical processing that occurs in the visual pathway compared to the strictly local processing of the retina and prCNN.

6.2 Step 2: Cross-Model Alignment

In the cross-model analysis, the overall alignment values across all metrics are relatively modest. This is most likely attributed to the inherent differences in objective functions: the prCNN is architecturally designed and explicitly trained to match biological neural activity in the retina, whereas DINOv2 is trained via self-supervision on image datasets, a task the biological retina does not perform.

Despite these modest absolute values, the relative patterns are highly informative. Embeddings and `norm1` layers are structurally the most similar to the prCNN. This occurs because the layer normalization operation is an affine rescaling that largely preserves upstream representations, acting essentially as a pass-through for the patch embeddings. The embeddings themselves are local linear projections of image patches, which is analogous to the local processing of the retina. Once the network reaches the attention layers and begins integrating information across the entire visual field, the representations diverge significantly from retinal processing.

The pass-through nature of the normalization layers also explains the sublayer matching patterns. Because `norm1` preserves the raw embedding, it is the best match for `conv0`, which itself is closer to the raw input. Conversely, the `attention` and `mlp` sublayers involve actual computation and spatial integration; thus, `conv1`, which performs slightly more sophisticated processing than `conv0`, is the relatively better match for those sublayers. Furthermore, `norm2` demonstrates an intriguing metric divergence: it has the highest Ridge R^2 because it contains both embedding and attention information, giving the regression more predictive material. However, it yields low CKA and RSA scores because the attention signal distorts the geometric structure away from retinal representations. This cleanly illustrates the difference between encoding-based and similarity-based alignment metrics.

7 Conclusion

In conclusion, this study demonstrates that self-supervised Vision Transformers like DINOv2 are a remarkably strong match for the primate cortical visual pathway. The V1 matching analysis highlights the significant computational transformation that occurs between the retina and V1: DINOv2's encoder 2 best captures V1 variance, and the fact that all DINOv2 layers up to encoder 2 vastly outperform all prCNN layers demonstrates that subcortical processing is substantial and is effectively captured by DINOv2.

Furthermore, this study locates the "retina" of DINOv2 effectively at its initial embeddings layer. The near-identity of the embeddings and `norm1` sublayers confirms that LayerNorm preserves the embedding representation. The embedding layer itself is a patch-level linear projection of the input—a local, relatively unprocessed transformation analogous to the retina's local receptive field processing. While this is a straightforward conclusion, it is mechanistically well-motivated given how locally the retina operates, highlighting the emergence of biologically plausible early visual representations in self-supervised transformers.

Future Directions: There are several avenues for future improvement. First of all, for the prCNN, instead of simply replicating a static input image 50 times across the time dimension, this study could jitter the images randomly on each frame according to a 2D Brownian motion to emulate fixational eye saccades. This better follows the methodology used by Gogliettino et al. [2] to capture natural temporal dynamics, and would provide a more biologically realistic input stream that more fully captures the temporal dynamics the model was designed to exploit. Secondly, recordings from subcortical parts of the visual pathway like the LGN or models of the LGN would be a valuable addition to the analysis to more comprehensively characterize the subcortical elements encoded by DINOv2.

References

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